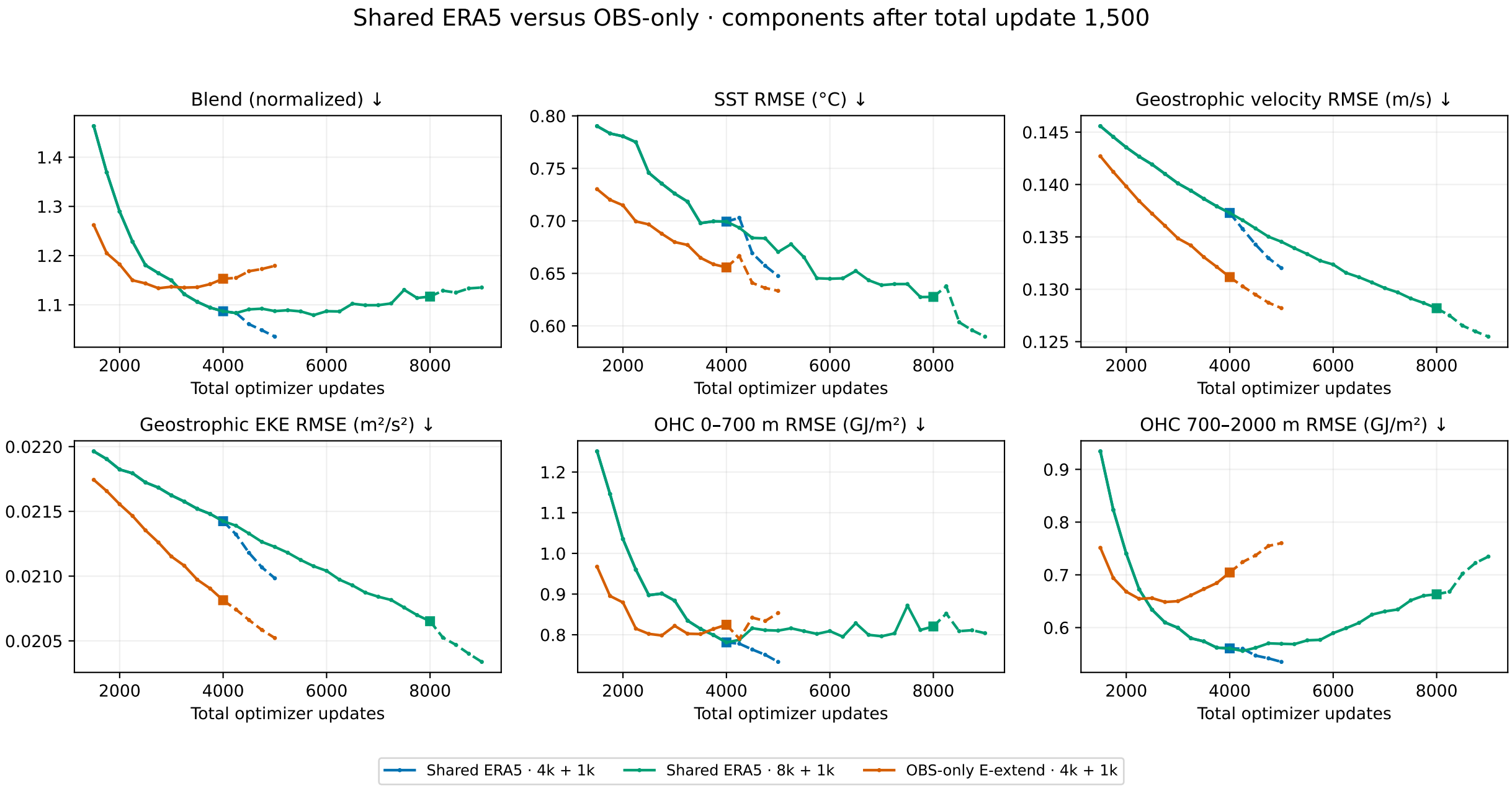
Shared ERA5 versus OBS-only · components after total update 1,500
Blend (normalized) ↓
SST RMSE (°C) ↓
Geostrophic velocity RMSE (m/s) ↓
Geostrophic EKE RMSE (m²/s²) ↓
OHC 0–700 m RMSE (GJ/m²) ↓
OHC 700–2000 m RMSE (GJ/m²) ↓
Total optimizer updates
Shared ERA5 · 4k + 1k
Shared ERA5 · 8k + 1k
OBS-only E-extend · 4k + 1k
Raw-field RMSE; surface leads 5/15/30 days, direct monthly OHC. Velocity/EKE derive from SSH, not the full-field X-ray velocity decodes.